LEXING ZHANG

Ph.D. Student in Transportation Engineering

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RESEARCH INTERESTS

Transportation network optimization and operations; bilevel and combinatorial optimization; branch-and-price-and-cut; network design; traffic flow and network control; mathematical programming and computational algorithms.

EDUCATION

University of Minnesota Twin Cities, Minneapolis, MN **Aug 2025–Present**
Ph.D. Student, Civil, Environmental, and Geo-Engineering (Transportation); Advisor: Prof. Michael W. Levin

National University of Singapore, Singapore **Aug 2024–May 2025**
Exchange Student, Department of Civil and Environmental Engineering

Chang'an University, Xi'an, China **Aug 2021–Jun 2025**
B.Eng., Transportation Engineering

ACADEMIC APPOINTMENTS

University of Minnesota Twin Cities 2025–Present
Graduate Research Assistant; Transportation network optimization and algorithm development, advised by Prof. Michael W. Levin

University of Minnesota Twin Cities 2026
Teaching Assistant, CEGE 3102

PEER-REVIEWED JOURNAL ARTICLES

1. **Zhang, L.**, Levin, M. W., and Nankali, M. (2027). *A bi-level optimization model for urban bicycle lane planning with a globally optimal branch-and-price-and-cut algorithm*. *Transportation Research Part E: Logistics and Transportation Review*, **217**, 105219. Published online in 2026. doi:10.1016/j.tre.2026.105219.

2. Feng, Y., Yao, E., Zhang, R., Chen, R., and **Zhang, L.** (2026). *Optimization of electric vehicle charging station layout considering mobile charging vehicles*. *Transportation Research Part D: Transport and Environment*, **159**, 105488. doi:10.1016/j.trd.2026.105488.

3. Ma, C., Shi, B., Cheng, Y., **Zhang, L.**, and Chen, K. (2026). *From lockdown to recovery: Modeling the long-term effects of COVID-19 on station-level ridership of Xi'an Metro*. *Transportation Research Record*, **2680**(1), 324–347. Published online in 2025. doi:10.1177/03611981251357005.

4. Wang, W., **Zhang, L.**, Yan, B., and Cheng, Y. (2025). *Development of a surrogate safety measure for evaluating rear-end collision risk perception*. *Transportation Research Record*, **2679**(6), 69–85. doi:10.1177/03611981241311574.

PEER-REVIEWED CONFERENCE PROCEEDINGS

1. Peng, B., **Zhang, L.**, Li, E., Feng, Y., Zhang, Y., and Zhong, L. (2024). *Free-flow tolling system for expressway with fusion of 5G communication and high-precision positioning technology*. In *Proceedings of SPIE: Fourth International Conference on Smart City Engineering and Public Transportation (SCEPT 2024)*, Vol. 13160, pp. 142–151. doi:10.1117/12.3030779.

HONORS AND AWARDS

- **CTS Travel Award**, University of Minnesota Center for Transportation Studies (2025).
- **National First Prize**, 18th Challenge Cup National College Students' Extracurricular Academic Science and Technology Works Competition (2023).
- **National Scholarship**, China (2022–2023 academic year).
- **Honorable Mention**, Mathematical Contest in Modeling (MCM), COMAP (2024).
- **Provincial First Prize**, National Undergraduate Mathematical Contest in Modeling, Shaanxi (2023).
- **Provincial First Prize**, Shaanxi College Students Transportation Technology Competition (2023).
- **Provincial Second Prizes**, Shaanxi College Students Transportation Technology Competition (2024).
- **University First Prize**, Chang'an University Challenge Cup Research Competition (2023).
- **Provincial Third Prize**, Shaanxi College Students Transportation Technology Competition (2024).
- **Class Academic Merit Scholarship**, Chang'an University (2022–2023 academic year).

TECHNICAL SKILLS

Programming and Software: Python, MATLAB, Gurobi, VISSIM.

Research Methods: Bilevel optimization, mixed-integer programming, branch-and-price-and-cut, column generation, network flow, shortest paths, mathematical modeling, computational experiments, numerical simulation, reinforcement learning, deep learning.